

University of Mines and Technology (UMaT), Tarkwa
Department of Mathematical Sciences

MPhil Thesis Proposal Presentation on
**A New Exponentiated Power Transform Generalised
(EPT-G) Family of Distributions: Properties,
Applications and Machine Learning Benchmarking**

By:
Donkoh, Benjamin Arkoh

Supervisors:
Assoc. Prof Joseph Acquah
Dr Abdulzeid Anafo

May 29, 2026



Presentation Outline

- 1 Research Background
- 2 Problem Statement
- 3 Literature Review
- 4 Research Questions
- 5 Research Objectives
- 6 Methods to be used
- 7 Expected Outcomes
- 8 Proposed Contributions to Science knowledge
- 9 Proposed Chapter Outline
- 10 Facilities and Resources to be Used
- 11 References



Research Background

Statistical distributions are fundamental for modelling real-world data in all fields of life. (Egamov *et al.*, 2024).

In the early 1890's, Pearson (1895) introduced the system of frequency curves; since then, researchers have continuously sought more flexible families of distributions. (Lahcene, 2013)

Traditional distributions like exponential, Weibull, and Gamma are widely used for modelling (Pizzutilo, 2012).



Research Background Cont'd

Machine learning (ML) is a branch of artificial intelligence that enables systems to learn patterns from data and make predictions (Khan *et al.*, 2025).

Many ML models rely on probabilistic assumptions and statistical distributions to model uncertainty (Tyralis and Pacharalampous, 2025).

Statistical distributions and ML models help evaluate predictive accuracy, robustness, and modelling flexibility. (Murlidhar and Sayed, 2024).



Problem Statement

Datasets in finance, epidemiology, and biomedical research often display complex features such as strong right-skewness, non-monotone hazard rates, and multimodal patterns. (Suleiman *et al.*, 2024).

The existing standard distributions, such as exponential, normal, and gamma, among others, do not really provide a good parametric fit to data sets that exhibit some of these characteristics (Anafo, 2019).



Problem Statement Cont'd

Some other distributions struggle to provide statistical properties, such as moments, entropy measures, and hazard rate functions. Examples of such families include the Kumaraswamy-G and beta-G families (Cordeiro and de Castro, 2011).

Machine learning research methods have brought new advanced density estimation techniques that provide an attractive alternative to traditional parametric distributions. (Sperti *et al.*, 2021)

However, there are also a few rigorous studies on benchmarks, especially in comparative studies between new statistical families and machine learning methods. (Lones, 2024).



Problem Statement Cont'd

The few existing studies on benchmarks are geared towards improving the ML models rather than the statistical distribution. (Lones, 2024)

Some other benchmarking practices are often deficient because of their unreliable train-test partitioning and the high level of variance in the results, which distorts the true performance of a particular model (Lones, 2024).

As a result, this research seeks to develop an EPT-G family of distribution, establish its theoretical properties, and demonstrate its practical use through real-world applications and compare machine learning with statistical distribution.



Literature Review

Citation	Topic	Methods	Gaps	Findings
Oluyede et al. (2020)	Exponentiated Generalized Power Series (EGPS) & EGL	- New family via EG power series - Derived properties (moments, entropy, order stats) - MLE + Monte Carlo	- No ML benchmarking - Limited computational efficiency analysis	- Flexible hazard shapes (increasing, decreasing, bathtub) - MLE consistent - Good real-life fit
Chettri et al. (2022)	New Alpha Power Exponential (NAPE)	- Applied NAPT to exponential - Derived properties (moments, PWMs, entropy, quantiles) - MLE + simulation	- Restricted to an exponential baseline - No ML benchmarking - Small sample Simulations	- Flexible hazard rates (monotone & non-monotone) - Outperformed classical models



Literature Review Cont'd

Anafo (2019)	New Alpha Power Transform (PT-G), applied to Weibull	- Proposed PT-G family - Derived properties - MLE + Monte Carlo - Guinea pig dataset application	- Limited to Weibull baseline - No ML benchmarking - No broad generalisation	- PT-Weibull flexible hazard shapes - MLE consistent - Better fit than competitors
Jia et al., (2020)	Applying Exponential Family Distribution to Generalised Extreme Learning Machine	-Extreme Learning Machine (ELM) neural network model. -Exponential Family Distribution framework. - MLE - Kernel Generalized Extreme Learning Machine (KGELM) for nonlinear modelling.	- Limited comparison with parametric statistical distributions. - Focus mainly on ML model improvement rather than distribution development. -No benchmarking with advanced statistical distribution families for density modelling	- GELM provides greater modelling flexibility. - The method improves predictive accuracy across datasets. -Kernel extension enhances performance on nonlinear problems.

Literature Review Cont'd

Elkalzah et al. (2025)	New Generalised Lomax Tangent Transformed Exponential (NGLTGE)	- Introduced Lomax tangent generalised transformation - Derived its properties - MLE + Monte Carlo simulations - Benchmarked against ML density estimators (Log-GMM, MAF)	- ML benchmarking limited to density estimation - Scalability across very large datasets has not been fully explored	- Highly flexible PDFs (unimodal, bimodal, skewed) - Hazard functions with diverse shapes - Outperformed classical rivals - Competitive with ML density estimators for heavy-tailed data
------------------------	--	---	--	--



Research Questions

- How can the EPT-G family of distributions be mathematically formulated, and What are its properties?
- What properties distinguish the EPT-Weibull distribution when derived as a member of the EPT-G family?
- How effectively does the EPT-Weibull distribution model fit into real-world datasets?
- How does the EPT-Weibull model benchmark against modern machine learning density estimators?



Research Objectives

The objectives of the study are to;

- develop the EPT-G family of distributions and its properties,
- introduce the EPT-Weibull as a special case and derive its properties,
- apply the EPT-Weibull to real datasets, and
- benchmark the EPT-Weibull against machine learning density estimators.



Methods to be use

The following are the methods to be used.

- The Cumulative Distribution Frequency (CDF) of the exponential family of distribution.
- The Weibull Distribution
- Monte Carlo Simulation and Estimation techniques (MLE, AD, CVM, OLS, WOLS and Percentile Estimation)
- The Log-Gaussian Mixture Model (Log-GMM) an the Affine NNormalizingFlow Model.



Expected Outcomes

The following are the expected outcomes.

- The EPT-G family and its properties would be developed,
- The EPT-Weibull as a special case and its properties would be formulated,
- The EPT-Weibull would be applied to real datasets, and
- The EPT-Weibull distribution would be benchmarked against machine-learning density estimators.



Proposed Contributions to Science knowledge

- The new EPT-G family would support complex survival and biomedical data, enabling more accurate risk stratification, prognosis, and healthcare resource planning.
- Benchmarking EPT-Weibull within ML pipelines would foster robust, transparent, and data-efficient AI for predictive maintenance
- The research would model environmental datasets with heavy tails and hazard flexibility, which would improve risk estimates for adaptation planning, disaster preparedness, and sustainability.



Proposed Chapter Outline

- Chapter 1 will be the introductory Chapter.
- Chapter 2 will be devoted to the literature review.
- Chapter 3 will focus on the development of the new EPT-G family of distributions and its various properties and the development of the EPT-Weibull distribution and its properties.
- Chapter 4 will emphasize on the results and discussion of the EPT-Weibull and ML density estimators performance.
- Chapter 5 will concentrate on conclusions and recommendations of the research.



Facilities and Resources to be Used

- University of Mines and Technology Library
- Personal Laptop
- Internet
- R Software



Reference

- Anafo, A.Y., (2019), *The New Alpha Power Transform: Properties and Applications*, MPhil Thesis, African Institute for Mathematical Sciences (AIMS), Ghana, 36 pp.
- Chettri, S., Das, B., Imliyangba, I. and Hazarika, P.J., (2022), "A Generalised Form of Power Transformation on Exponential Family of Distribution with Properties and Application," *Pakistan Journal of Statistics and Operations Research*, pp. 511-535.
- Cordeiro, G. M., Ortega, E. M. M. and Ramires, T. G., (2015). A New Generalised Weibull Family of Distributions: Mathematical Properties and Applications. *Journal of Statistical Distributions and Applications*, Vol. 2, No. 1.
- Egamov, M., Sharipov, E., Shodiev, S. and Abdiraxmonov, A., (2024), Application of Mathematical Modelling in Probability Theory and Mathematical Statistics. *E3S Web of Conferences*, 583 pp.



Reference Cont'd

- Elkalzah, B., Oguadimma, E.E., Obieke, V.C. and Michael, C., (2025), "A New Family of Generalized Distributions, with Applications and Benchmarking against Machine Learning Models," *International Journal*, Vol. 14, 887 pp.
- Jia, Y., Kwong, S. and Wang, R., (2018), "Applying exponential family distribution to generalized extreme learning machine," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, Vol. 50, No.5, pp.1794-1804.
- Khan, F., Ojha, P., Ansari, G. F. (2025), "Machine Learning (ML) and Artificial Intelligence (AI) Approaches to Unstructured Data", *Pure and Applied Mathematics Journal*, Vol. 14, No. 5, pp 114–119.
- Lahcene, B., (2013), "On Pearson Families of Distributions and its Applications," *African Journal of Mathematics and Computer Science Research*, Vol. 6, No. 5, pp. 108-117.



Reference Cont'd

- Lones M. A., (2024), "Avoiding Common Machine Learning Pitfalls", *Patterns*, Vol. 5, No. 10, pp. 2666-3899
- Murlidhar, P.S. and Sayed, S.A., (2024), "Uncertainty Quantification in Machine Learning Predictions: A Bayesian Statistical Perspective", *International Journal of Engineering Science and Humanities*, Vol. 14, No. 3, pp. 26-37.
- Oluyede, B.O., Mashabe, B., Fagbamigbe, A., Makubate, B. and Wanduku, D., (2020), "The Exponentiated Generalized Power Series: Family Of Distributions: Theory, Properties And Applications," *Heliyon*, Vol. 6, No. 8.
- Pearson, K., (1895), "X. Contributions to the Mathematical Theory of Evolution. II. Skew Variation in Homogeneous Material", *Philosophical Transactions of the Royal Society of London, Series A: Containing Papers of a Mathematical or Physical Character*, Vol. 186, pp.343-414.



Reference Cont'd

- Pizzutilo, F., (2012), "Use of the Pearson System of Frequency Curves for the Analysis of Stock Return Distributions: Evidence and Implications for the Italian Market", *Economics Bulletin*, Vol. 32, pp. 272-281.
- Sperti, M., D'Ascenzo, F., Navarini, L., Di Benedetto, G., Afeltra, A., Giacomelli, R., & Deriu, M. A., (2021), "Machine learning approaches as an alternative to traditional statistical methods in cardiovascular risk prediction", *Biomedical Science and Engineering*, Vol. 2, No. 1, pp. 109-110.
- Suleiman, A. A., Daud, A., Ishaq A. I., Kayid, M., Sokkalingam, R., Hamed, Y., Othman, M., Nagarjuna, V. B. V., Elgarhy, M., (2024), "A New Weibull distribution for Modelling Complex Biomedical Data", *Journal of Radiation Research and Applied Sciences*, Vol. 17, No. 4, 101190 pp.



Reference Cont'd

- Tyrallis, H. and Papacharalampous, G., (2024), "A review of predictive uncertainty estimation with machine learning", *Artificial Intelligence Review*, Vol. 57, No. 4, p.94



THANK
YOU

